

Characterization of the mechanical properties of a dermal equivalent compared with human skin *in vivo* by indentation and static friction tests

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Background/aims: The study of changes in skin structure with age is becoming all the more important with the increase in life. The atrophy that occurs during aging is accompanied by more profound changes, with a loss of organization within the elastic collagen network and alterations in the basal elements. The aim of this study is to present a method to determine the mechanical properties of total human skin *in vivo* compared with dermal equivalents (DEs) using indentation and static friction tests.

Methods: A new bio-tribometer working at a low contact pressure for the characterization the mechanical properties of the skin has been developed. This device, based on indentation and static friction tests, also allows to characterize the skin *in vivo* and reconstructed DEs in a wide range of light contact forces, stress and strain.

Results: This original bio-tribometer shows the ability to assess the skin elasticity and friction force in a wide range of light normal load (0.5–2 g) and low contact pressure (0.5–2 kPa). The results obtained by this approach show identical values of the Young's modulus E^* and the shear modulus G^* of six DEs obtained from a 62-year-old subject ($E^* = 8.5 \pm 1.74$ kPa and $G^* = 3.3 \pm 0.46$ kPa) and *in vivo* total skin of 20 subjects aged 55 to 70 years ($E^* = 8.3 \pm 2.1$ kPa, $G^* = 2.8 \pm 0.8$ kPa).

Key words: indentation – stiffness – friction – dermal equivalents – skin *in vivo*

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TO ASSESS the mechanical properties of the skin different noninvasive techniques, directly inspired by materials science, have been developed. The most commonly used methods are based on the measurement of suction, which consists in applying a negative pressure to the skin (1, 2), torsion (3, 4) and traction, by applying a linear displacement in the horizontal plane of the skin (5). Other experimental methods can be used to measure the mechanical behaviour of skin such as a tensile test, where two tabs are glued to the skin and driven apart from each other (6–8), and elastic wave propagation (9, 10). In the literature, the Young's modulus of the skin, E^* , varies between 0.42 and 0.85 MPa for torsion tests (11), 4.6 and 20 MPa for the extension tests (12) and between 0.05 and 0.15 MPa for the suction tests (1, 2).

However, these methods have several disadvantages. (1) the prepressure of the skin, which depends on the user and the manner in which

pressure of contact is applied between the probe and the skin. (2) The solicited layer depends on the geometry of the probe. (3) All these methods are used to characterize the skin rheology by descriptive parameters *in vivo*.

In order to develop a noninvasive metrology to characterize the mechanical behavior of both DE and complete skin *in vivo*, a specific indentation/friction device has been designed (13–15). The metrological equipment is an original tribological system based on the measurement of static and sliding contact forces between a rigid spherical probe and a soft material. During the indentation test, the technique consists in monitoring the penetration of the indenter into the surface while displacement δ and the normal force F_N are continuously measured with a high sensitivity.

The measurement of complete penetration δ at the maximum of indentation force $F_{\max}$ allows to determine the radius of contact ' a ' and the mean contact pressure by applying the Hertz theory.

The determination of the slope from the maximum of the unloading curve allows to quantify the contact stiffness ($K_N = \frac{dF}{d\delta}$) and the reduced Young's modulus E^* .

When the probe is moved along the surface over a certain length with a sliding velocity in the range of 100 to 1000 $\mu\text{m/s}$, the friction force F_T of a surface can be characterized in two distinct phases: in the static phase, the lateral force increases linearly to reach a maximum that corresponds to the static limit friction force. During this static resistance phase, we can quantify the lateral stiffness K_T (N/m) and the shear modulus G^* . In the sliding phase, the simultaneous measurement of the normal and friction force F_N , F_T , can be used to determine the dynamic friction coefficients, μ_d . During this sliding phase, a discontinuous phenomenon called stick-slip, due to compression and distortion of the skin, occurs during the sliding process.

Materials and Method

The bio-tribometer used in this study uses a fully automated tribological configuration based on the metrology of static and linear sliding contact between a spherical indenter and a soft material (Fig. 1). This tribometer is designed to measure the indentation cycle and frictional forces in a wide range of weak normal forces and contact pressures, varying, respectively, between 5 and 100 mN and 0.5 to 2 kPa. The automated system controls the applied normal load, penetration depth, normal and lateral velocity. A smooth steel spherical indenter is selected as a probe to facilitate theoretical modeling.

The metrology developed consists of the control of the normal or the lateral displacements

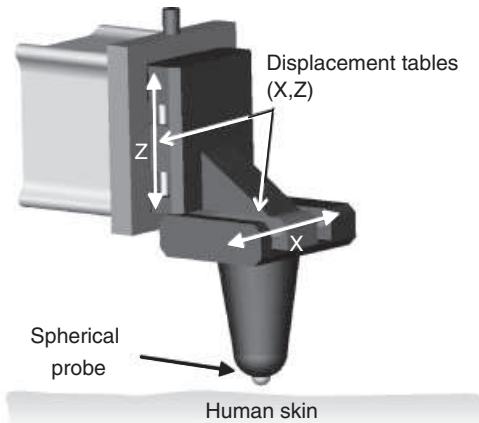


Fig. 1. Skin bio-tribometer.

of the probe. The electrical signal of measured forces is converted to digital information and collected by the computer.

Validation on Inert Material

The developed device has been tested and validated on a pure elastic material and a visco-elastic elastomer. The pure elasticity was tested on helicoïdal springs with various known stiffness. Two springs were used, with 300 and 570 N/m stiffness. The results presented in Fig. 2 show the device validity.

To validate the indentation test on a visco-elastic material, we analyzed the mechanical behavior of a standard elastomer. The reduced Young's modulus E^* of our elastomer was 2 ± 0.2 MPa (test conducted in nano-indentation). Figure 3 shows an indentation cycle on a standard elastomer with a normal load of 20 mN. We can note the presence of adhesive forces at the end of the indentation cycle. This phenomenon results in negative efforts. The Young's modulus value determined in the unloading phase is about 2.4 ± 0.4 MPa, which is in accordance with the value measured with a nano-indentation test.

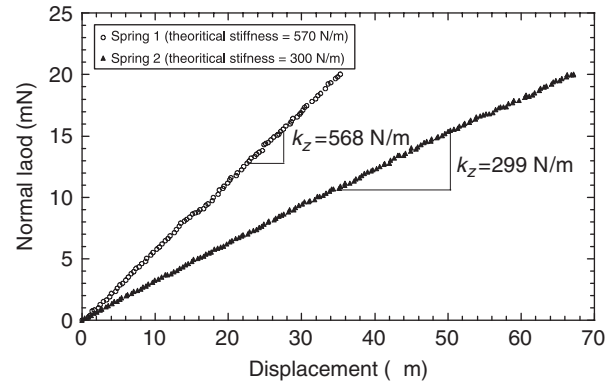


Fig. 2. Measurement of stiffness of two helicoïdal springs (Spring 1: $K_z = 568 \pm 0.003$ N/m, Spring 2: $K_z = 299 \pm 0.003$ N/m).

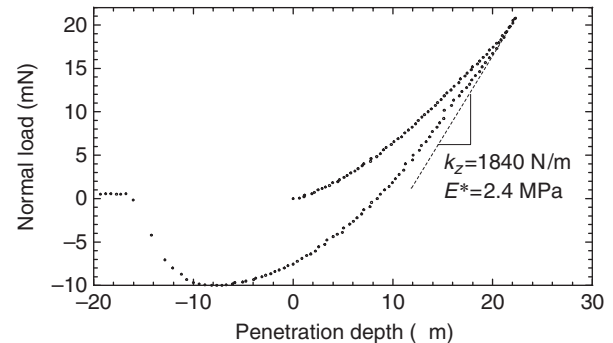


Fig. 3. Indentation test for elastomer in ambient air.

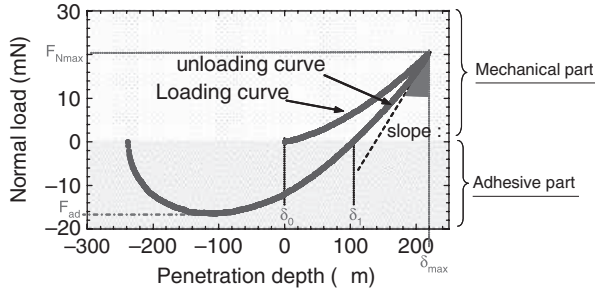


Fig. 4. Loading-unloading curve during the indentation test on soft adhesive viscoelastic elastomer for indentation speed $10 \mu\text{m/s}$. The indentation curve is divided in two parts: mechanical part (bulk properties) and adhesive part (surface properties).

Assessment of the Indentation Curve and Determination of the Stiffness and Young's Modulus

In the loading-unloading cycle (hysteresis cycle), not only the elastic component but also the viscous properties can be assessed. When the given normal pressure is achieved, during the unloading phase, the indenter is returned to the starting position with the same velocity. A hypothetical set of continuous load-displacement data is presented in Fig. 4.

Some important parameters are the peak load and displacement, $F_{N\max}$, $\delta_{\max}$ and the slope of the initial portion of the unloading curve, $K_N = \frac{dF}{d\delta}$. Note that K_N has the dimensions of force per unit distance, and so it is also known as the elastic stiffness of the contact. For any axisymmetric indenter, the reduced elastic modulus is obtained from (16, 17):

$$E^* = \frac{\sqrt{\pi} K_N}{2 \sqrt{A}} \quad (1)$$

where E^* is the reduced Young's modulus:

$$\frac{1}{E^*} = \frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \quad (2)$$

with E_1 and E_2 , respectively, being the Young's modulus of human skin and the Young's modulus of the steel spherical indenter and ν the Poisson's ratio; hence, $E_1 \gg E_2$, and the reduced Young's modulus is $\frac{1}{E^*} = \frac{1 - \nu_2^2}{E_2}$. K_N is the stiffness of the contact and A is the projected normal area. The projected normal contact area is: $A = \pi a^2$, where a is the Hertzian's contact radius and is given by

$$a = \sqrt{\delta R} \quad (3)$$

where R are, respectively, the penetration depth and the radius of curvature of the spherical indenter.

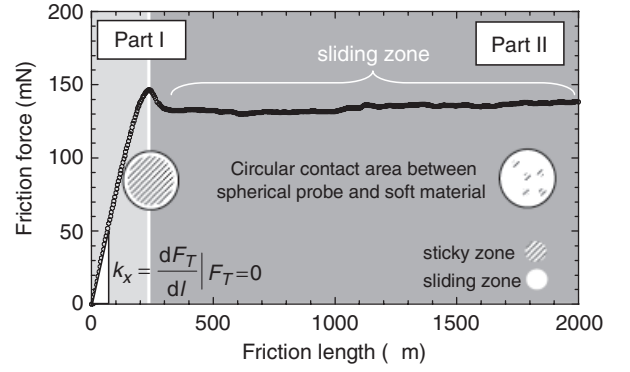


Fig. 5. Variation of the friction force vs. friction length onto the elastomer used for the indentation test. The friction curve is divided in two different zones: sticky (or static) zone (part I) and sliding zone (part II). The applied normal load is $F_N = 20 \text{ mN}$ and the sliding velocity is $v = 200 \mu\text{m/s}$. The test was conducted with a smooth spherical steel probe with a radius of curvature $R = 2.25 \text{ mm}$.

Friction Test

The friction test consists of applying onto the surface of the material a constant normal load and a constant sliding velocity and measuring the resulting friction force, F_T . Classically, the friction, F_T , can be divided into two parts (Fig. 5). The first part (part I) corresponds to the sticky zone (or quasi-static zone) between the probe and the material; there is no relative displacement between the slider and the material (18). The quasi-static zone reflects the tangential elastic properties of the material. In this zone, the lateral stiffness, k_x , is defined as the slope of the initial portion of the friction curve (Fig. 2): $k_x = \frac{dF_T}{dl} |_{F_T=0}$ and the shear modulus is linked to the lateral stiffness by (19):

$$G^* = \frac{k_x}{8a} = \frac{\sqrt{\pi} k_x}{8 \sqrt{A}} \quad (4)$$

where A is the projected contact area. G^* is defined as:

$$\frac{1}{G^*} = \frac{2 - \nu_1}{G_1} + \frac{2 - \nu_2}{G_2} \quad (5)$$

with G_1 , ν_1 and G_2 , ν_2 being the respective shear modulus and Poisson's ratio of the steel spherical probe and the indented material. In this study, G_1 (steel) $\gg G_2$ (soft material); it can be assumed that G^* is the reduced shear modulus of the tested material,

$$\frac{1}{G^*} \approx \frac{2 - \nu_2}{G_2} \quad (6)$$

As already described before, the penetration depth, δ , is related to the contact radius, a , of the projected contact area ($A = \pi a^2$) by $\delta = \frac{a^2}{R}$; thus, Eq. (3) becomes

$$G^* = \frac{k_x}{8\sqrt{\delta R}} \quad (7)$$

The second part of the curve (part II) corresponds to the sliding zone between the probe and the material. In the sliding zone, the friction coefficient, μ , is determined as the ratio between friction force, F_T , and normal load, F_N :

$$\mu = \frac{F_T}{F_N} \quad (8)$$

Application to Study the Mechanical Properties of the Skin

The skin is a composite material stratified into three layers with different mechanical properties. The superficial layer, the 15 μm thick *stratum corneum*, is very firm but pliable and naturally wrinkled; underneath, the less rigid 50 μm thick viable epidermis is also wrinkled and the underlying third layer, the 1 mm thick dermis, is elastic and naturally contracted. These three layers lie on the hypodermis (or *sucutis*), whose thickness, often filled with adipose tissue, can range from 1 mm to more than 5 cm and behaves like a cushion. Various structures with specific mechanical roles are found in each layer. The major mechanical component is the dermis. Similar to all connective tissues, it is a felting of collagen fibers immersed in a viscous fluid, called ground substance, rich in glycoproteins. The dermis has two main functions: to maintain the spatial organization of underlying tissues and to resist external mechanical stress.

With the intention of characterizing the mechanical behavior of the dermis and the complete skin, we studied experimentally the behavior of the DE and the complete skin of a population of the same age as the subject, whose DE was cultivated.

The purpose is to identify the contribution of the dermis during a compressive test of the complete skin in the normal and the horizontal direction by indentation and static friction tests.

Mechanical Properties of DE

Dermal substrate

Briefly, 1.5% bovine collagens (95% type I, 5% type III) were solubilized in 0.025 M acetic acid. Chitosan from crab-deacetylated chitin was dissolved at 0.7% in 0.084 M acetic acid. Chondroitin-

4, sulfate from bovine cartilage was prepared at 2.5% in deionized water. All these ingredients were mixed to obtain the following final composition in dry matter: 72% collagen, 20% chitosan and 8% chondroitin-4, sulfate. This final preparation was poured into Snapwell inserts (Corning Life Sciences, New York, NY, USA) at 400 mg/cm², and then freeze-dried overnight (−40 °C/+37 °C, 0.1 mbar) and sterilized by γ irradiation (10 kGy).

Human dermal fibroblast isolation

Fibroblasts were isolated from a number of human adult skin specimens. Fibroblasts were grown in fibroblast medium consisting of DMEM (Invitrogen, Carlsbad, CA, USA) supplemented with 10% newborn calf serum (HyClone, Logan, UT, USA), 25 mg/L gentamycin, 100,000 UI/L penicillin and 1 mg/L amphotericin B, and were used between passages 5 and 7. For each experiment, the DEs used were all constituted with the same batches of keratinocytes and fibroblasts.

DE

DEs were prepared by adding a suspension of 2×10^5 human fibroblasts on top of each dermal substrate rehydrated previously with DMEM. All equivalents were cultured for 29 days in fibroblast medium containing 1 mM ascorbic acid 2-phosphate (Sigma, Saint-Quentin Fallavier, France). The culture medium was renewed every day. Culture supernatants were harvested on days 7, 14 and 21 for cytokine measurements (Fig. 6).

The mechanical properties of six DEs of a 62-year-old subject were assessed by indentation and static friction tests. Measurements were

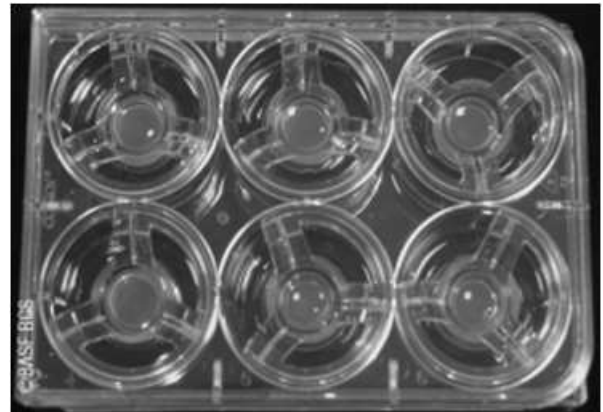


Fig. 6. In vitro 3D DEs after 21 days of culture.

made after 15 min of stabilization in a room with controlled temperature and humidity (22 °C and RH 50%). At first, the indentation test was performed with a light normal load of 5 mN (0.5 g), which corresponds to a mean contact pressure of ($\cong 1$ kPa). The indentation speed used was about 20 $\mu\text{m/s}$. For the same load and contact pressure, the friction test was performed for a length of 1 mm with a lateral speed of 50 $\mu\text{m/s}$. Figures 7 and 8 show the mean results of the indentation and friction force of DEs. After the characterization of the experimental curves and statistical analysis, we obtained a mean value of Young's modulus $E^* = 8.54 \pm 1.74$ kPa and a mean shear modulus $G^* = 3.3 \pm 0.46$ kPa. These results show the anisotropic behavior of DE elasticity in the normal and horizontal directions with a characteristic parameter of anisotropy $G^*/E^* = 0.388 \approx 3/8$.

These two elastic moduli in both directions as well as the anisotropy parameter will be compared with those of the complete skin *in vivo* in the same experiment.

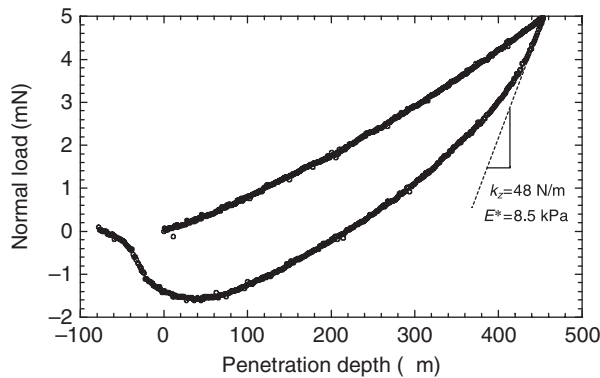


Fig. 7. Indentation test on dermal equivalents (N = 6), accomplished for a normal pressure of 1 kPa (donor 62 years old).

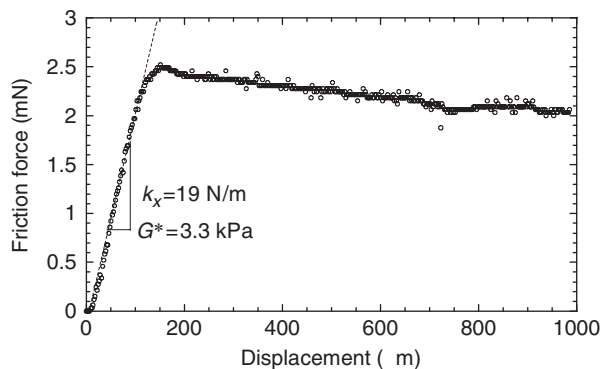


Fig. 8. Friction force on dermal equivalent. (N = 6). Test accomplished for a normal pressure of 1 kPa and a normal strain of 9% for a length of 1 mm and a sliding speed of 50 $\mu\text{m/s}$. (donor 62 years old).



Fig. 9. Indentation and friction device of skin (bio-tribometer).

In Vivo Measurement

Indentation and friction tests of total skin *in vivo* were performed on 20 healthy women volunteers, aged from 55 to 70 years. All volunteers gave their informed consent; they were free of pathological findings on their arms and were not using any topical agents on the tested area for 72 h.

Measurements were performed after 30 min of stabilization in a room with controlled temperature and humidity (22 °C and RH 50%). Subjects were seated in a medical chair with a specially designed armrest. The right forearm was fixed on the wrist to minimize movement, Fig. 9.

Three measurements with complete repositioning of the probe were performed during the indentation and friction tests. The duration of an indentation test is of the order of 5 s, while the friction test takes 10 s, depending on the length and velocity of friction chosen. The average of these three measurements is used in normal and lateral stiffness calculations.

All tests were performed at a force of 20 mN with a 1000 μm penetration depth, which corresponds to a mean contact Hertzian pressure of contact of 1 kPa.

After the characterization of the experimental curves (Figs 10 and 11) and statistical analysis, we obtained a mean value of Young's modulus $E^* = 8.3 \pm 2.1$ kPa and a mean shear modulus $G^* = 2.8 \pm 0.46$ kPa. These results indicate an anisotropic behavior in terms of total skin elasticity in the normal and horizontal directions with a characteristic parameter of anisotropy $G^*/E^* = 0.337$.

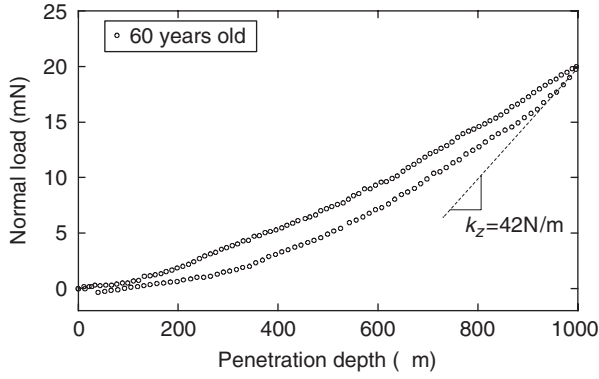


Fig. 10. Indentation curve of volar forearm *in vivo*: $E^* = 8, 3 \text{ kPa} \pm 2.1 \text{ kPa}$.

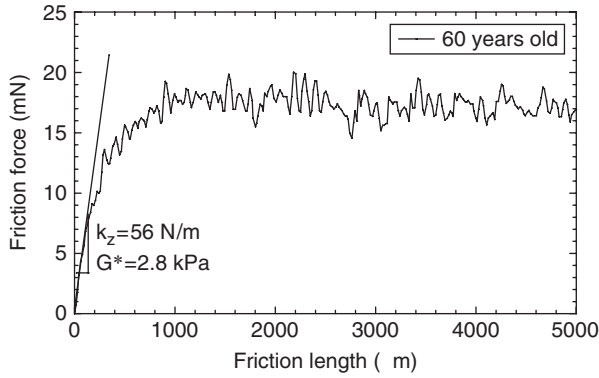


Fig. 11. Friction force of skin.

TABLE 1. Mechanical parameters of dermal equivalent and total skin *in vivo*

Mechanical parameters	Dermal equivalents	Skin <i>in vivo</i>
E^*	8.5 KPa	8.3 KPa
G^*	3.3 KPa	2.8 KPa
G^*/E^*	0.388	0.337

In Table 1, we compare the results of the mechanical characterization of DE and total skin *in vivo*. These results show a huge similarity between DEs and total skin *in vivo* under a compressive stress of 1 kPa. This similarity in stiffness between DEs and total skin *in vivo* shows that the dermis acts as the place of resistance to external stress, and, during the compressive study using our bio-tribometer, we found a significant contribution of the dermis in the visco-elastic reaction to the indentation and static friction test *in vivo*.

As shown in Fig. 12, on comparing the Young's modulus of the total skin and DEs in a large spectrum of materials, we can note that the human skin is part of ultrasoft materials, where its elasticity is provided by collagen fibers (which

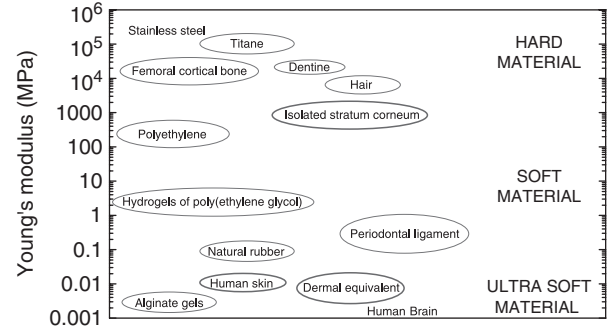


Fig. 12. Young's modulus of various materials.

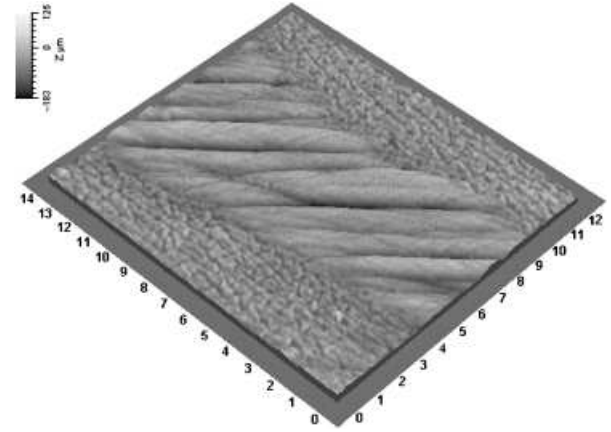


Fig. 13. Stretch marks.

provide skin firmness) and elastin fibers (which provide softness). These elastic fibers, produced by the organism from the fetal stadium until the age of 15–20 years, and that constitute a kind of 'elastic capital' to crack and disintegrate progressively. A partial failure of the skin elasticity is found in hypertrophic scars, keloids and stretch marks, Fig. 13. As described by Piérard (20), the permanent tension of the reticular dermis exerted by elastic fibers generates the folding of non-elastic overlying structures (viable epidermis and *stratum corneum*). This is the origin of the skin surface relief, which appears first at the deeper level of the subpapillary dermis. During chronological aging, the density and orientation of the elastic fibers change and give rise to a marked anisotropic topography, Fig. 14, leading to a significant loss of skin elasticity, Fig. 15.

Conclusions

The development of new cosmetic formulations requires a precise assessment of their safety and efficacy. Today, legislation imposes increasing measures of safety and claims substantiation as

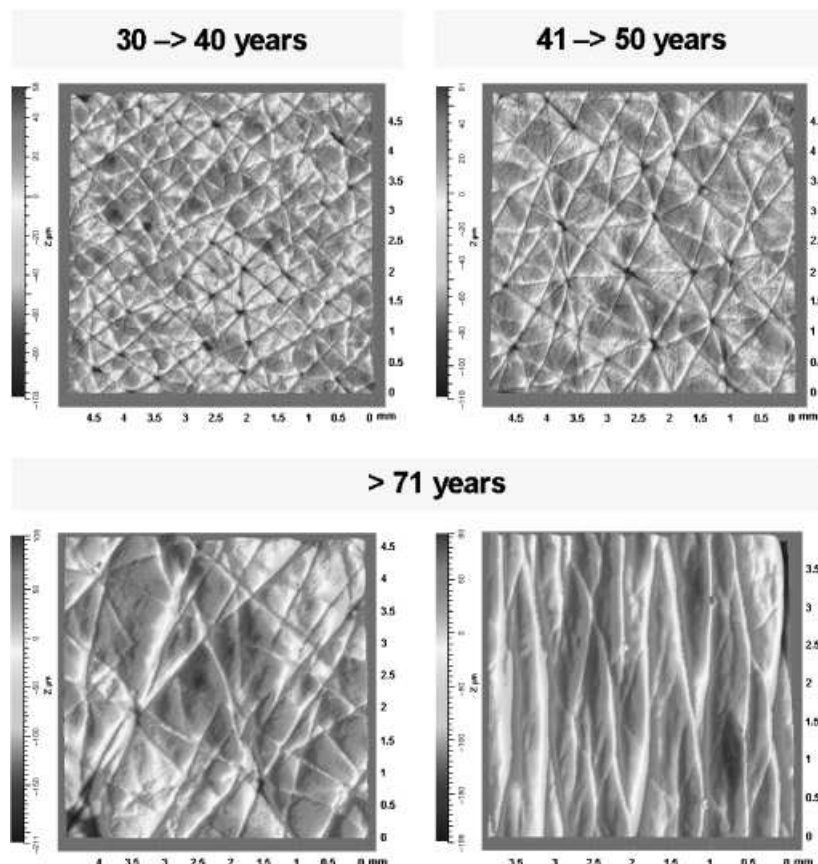


Fig. 14. Morphology of skin topography during ageing assessed by laser microscopy.

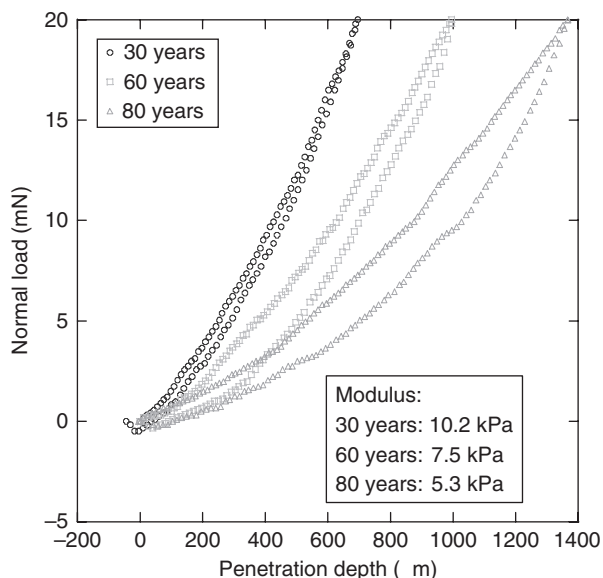


Fig. 15. Indentation test of skin during ageing.

well as limitation of animal use for such testing. Subsequently, assessment protocols now focus on *in vivo* human volunteer tests and *in vitro* methods. This has led to the development of new *in vitro* models to predict either the irritancy

potential of substances or their cosmetic efficiency (21). For the last 10 years, most *in vitro* studies demonstrating the biological effects of cosmetic actives have been performed on monolayered cultures. Although all the cutaneous cells can be maintained in a monolayered culture, this conventional model is oversimplified. Indeed, cells that do not reproduce the *in vivo* situation in terms of the lack of cell-cell and cell-ECM interactions are isolated. The development of three-dimensional models such as Mimederm[®] (BASF BCS, Lyon, France) is more similar to normal human skin and allows both these kinds of interactions and topical application of the tested cosmetics, mimicking normal conditions for use of the product in its final form (22).

These models have been designed to meet the research and testing needs of the cosmetics, pharmaceutical and chemical industries and for studies of skin cellular and molecular biology. Therefore, these three-dimensional models are relevant tools for discovering new cellular targets during the aging process for example (comparison between young and old donors) and also for screening new potential active compounds that

can present significant efficacy in the new identified cellular targets (22, 23).

The successful application of tissue-engineered skin models requires that the morphological and ultrastructural organization mimic the normal skin as closely as possible. Moreover, this model can be developed by adding new cell types such as dermal dendritic cells or adipocytes to test new classes of cosmetic ingredients.

The results obtained with the use of our biotribometer are in perfect agreement with recent studies concerning the determination of the Young's modulus by indentation. Jachowicz et al. (24) found for corresponding values for forearm and facial skin of 10 panelists that the Young's modulus is in the range of 7–33 kPa. Delalleau et al. (25) studied a coupling approach, based on the inverse method of the indentation test with a numerical simulation, modeled by finite elements using the SYSTUS software. The results show that for an indentation depth of 50–600 μm , the global stabilization yields a Young's modulus of skin about 5.67 kPa and a Poisson coefficient of 0.48. In order to determine contributions of the dermis and hypodermis during a compressive test, Pailler-Mattei et al. (26) developed a mechanical model by considering two layers. This approach allowed to account for the observed decrease of properties of the skin up to a penetration depth equal to its thickness. Realistic values for the Young's modulus of each layer that constitutes the skin are thus obtained. The model gives an average value of the skin Young's modulus between 4.5 and 8 kPa for the volar forearm zone.

The comparison between the Young modulus of the skin *in vivo* and the DEs shows a high similarity, which may indicate that the characterization of the skin *in vivo* by a compressive test in the range of normal load (0.5–2 g) and contact pressure (0.5–2 kPa) is dependent on the collagen and elastin fibers of the dermis, which provide skin elasticity.

References

- Diridollou S, Patat F, Gens F, Vaillant L, Black D, Lagarde JM, Gall Y, Berson M. In vivo model of the mechanical properties of the human skin under suction. *Skin Res Technol* 2000; 6: 214–221.
- Hendriks FM, Brokken D, Van Eemeren JTWM, Oomens CWJ, Baaijens FPT, Horsten JBAM. A numerical-experimental method to characterize the non linear mechanical behaviour of human skin. *Skin Res Technol* 2003; 9: 274–283.
- Sanders R. Torsional elasticity of human skin in vivo. *Pflugers Archiv Eur J Physiol* 1973; 342: 255–260.
- Berardesca E, de Riga J, Leveque JL, Maibach HI. In vivo biophysical characterisation of skin physiological differences in races. *Dermatologica* 1991; 182: 89–93.
- Sugihara T, Ohura T, Homma K, Igawa HH. The extensibility in human skin: variation according to age and site. *Brit J Plastic Surg* 1991; 44: 418–422.
- Alexander H, Cook TH. Accounting for natural tension in the mechanical testing of human skin. *J Invest Dermatol* 1977; 69: 310–314.
- Del Prete Z, Antonucci S, Hoffman AH, Grigg P. Viscoelastic properties of skin in Mov-13 and Tsk mince. *J Biomech* 2004; 37: 1491–1497.
- Ozyazgan I, Liman N, Dursun N, Gunes I. The effects of ovariectomy on the mechanical properties of skin in rats. *Maturitas* 2002; 43: 65–74.
- Moran CM, Bush NL, Bamber JC. Ultrasonic propagation properties of excised human skin. *Ultrasound Med Biol* 1995; 21: 1177–1190.
- Pan L, Zan L, Foster FS. Ultrasonic and viscoelastic properties of skin under transverse mechanical stress in vitro. *Ultrasound Med Biol* 1998; 24: 995–1007.
- Agache PG, Monneur C, Leveque J-L, De Riga J. Mechanical properties and Young's modulus of human skin in vivo. *Archives Dermatol Res* 1980; 269: 127–133.
- Manschot JF, Brakkee AJ. The measurement and modelling of the mechanical properties of human skin in vivo I. The measurement. *J Biomech* 1986; 19: 511–515.
- Zahouani H, Pailler-Mattei C, Vargiolu R, Abellan MA. Assessment of the elasticity and tactile properties of the human skin surface by tribological tests. *Edinburgh: 22nd IFSCC Congress, 2002; (Proceeding)*.
- Pailler-Mattei C, Zahouani H. Study of adhesion forces and mechanical properties of human skin in vivo. *J Adhesion Sci Technol* 2004; 18: 1739–1758.
- Pailler-Mattei C, Zahouani H. Analysis of adhesive behaviour of human skin in vivo by indentation test. *Tribol Int* 2006; 39: 12–21.
- Loubet JL, Georges JM, Marchesini O, Meille G. Vickers indentation curves of magnesium oxide (MgO). *J Tribol* 1984; 106: 43–48.
- Sneddon IN. The relation between load and penetration in the axiometric Boussinesq problem for a punch of arbitrary profile. *Int J Eng Sci* 1965; 3: 47–57.
- Mindlin RD. Compliance of elastic bodies in contact. *J Appl Mech* 1949; 6: 259–268.
- Johnson KL. *Contact mechanics*. England: Cambridge University Press, 2001.
- Pierrard GE. EEMCO guidance to the in vivo assessment of tensile functional properties of the skin. Part 1: relevance to the structures and ageing of the skin and subcutaneous tissues. *Skin Pharmacol Appl Skin Physiol* 1999; 12: 352–362.
- Black AF, Bouez C, Perrier E, Schlotmann K, Chapuis F, Damour O. Optimization and characterization of an engineered human skin equivalent. *Tissue Eng* 2005; 11: 723–733.
- André V, Grenier S, Pivard F, Perrier E. 3D cellular models: a powerful access to cutaneous physiology and to innovative developments of cosmetic active compounds. *Pathol Biol* 2005; 53: 618–626.
- Cenizo V, André V, Reymermier C, Sommer P, Damour O, Perrier E. LOXL as a target to increase the elastin content in adult skin: a dill extract induces the LOXL gene expression. *Exp Dermatol* 2006; 12: 1–10.

24. Jachowicz J, McMullen R, Prettypaul D. Indentometric analysis of *in vivo* skin and comparison with artificial skin models. *Skin Res Technol* 2007; 13: 299–309.
25. Delalleau A, Josse G, Lagarde JM, Zahouani H, Bergheau JM. Characterization of the mechanical properties of skin by inverse analysis combined with the indentation test. *J Biomech* 2006; 39: 1603–1610.
26. Pailler-Mattei C, Bec S, Zahouani H. In vivo measurements of the elastic mechanical properties of human skin by indentation tests. *Med Eng Phys* 2008; 30: 599–606.

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